Lezione 29/04/26 3 ore

Esercizio

Banca con due sportelli

N_i = # numero clienti in coda indipendenti

$\mathbb{E}[N_1] = 8 \qquad \mathbb{E}[N_2] = 4$

$N = N_1 + N_2$

$\mathbb{P}[N = 9] \qquad \mathbb{P}[N_2 = k \mid N = n]$

$N \sim \text{Poisson}(12)$

$$\mathbb{P}[N = 9] = e^{-\lambda} \frac{\lambda^k}{k!} \simeq 8.7\%$$

$$\mathbb{P}[N_2 = k \mid N = n] = \frac{\mathbb{P}[N_1 + N_2 = n \mid N_2 = k]}{\mathbb{P}[N = n]}$$

$$= \frac{\mathbb{P}[N_1 = n-k]\,\mathbb{P}[N_2 = k]}{\mathbb{P}[N = n]}$$

$$= \frac{e^{-\lambda_1} \frac{\lambda_1^{n-k}}{(n-k)!} \; e^{-\lambda_2} \frac{\lambda_2^{k}}{k!}}{e^{-(\lambda_1+\lambda_2)} \frac{(\lambda_1+\lambda_2)^n}{n!}}$$

$$= \binom{n}{k} \left(\frac{\lambda_2}{\lambda_1+\lambda_2}\right)^k \left(\frac{\lambda_1}{\lambda_1+\lambda_2}\right)^{n-k}$$

$p = \frac{\lambda_2}{\lambda_1+\lambda_2}$

$= \mathbb{P}[S = k] \qquad S \sim \text{Bin}(n, p)$

- DISTRIBUZIONE IPERGEOMETRICA

$$\tilde{S}_n \sim \text{Hyper}(r, b, n) \qquad 1 \le n \le r+b$$

MODELLO : urne con r palline rosse b bianche

n palline estratte senza rimpiazzo

a) $\tilde{S}_n$ = # palline rosse estratte

b) $\tilde{S}_n \in \{0, 1, \dots, n\}$

c) $$\mathbb{P}[\tilde{S}_n = k] = \frac{\binom{r}{k}\binom{b}{n-k}}{\binom{r+b}{n}}$$

$k \le r$

$n-k \le b$

d) $$\mathbb{E}[\tilde{S}_n] = n\,\frac{r}{r+b} \qquad \text{Var}[\tilde{S}_n] = n\,\frac{r}{r+b}\,\frac{b}{r+b}\,\frac{r+b-n}{r+b-1}$$

f) $$\mathbb{P}[\tilde{S}_n = k] \simeq \mathbb{P}[S_n = k]$$

$r \gg 1$, $b \gg 1$, $p = \frac{r}{r+b}$

Esercizio

Una ditta produce 1000 lampadine e' ora di cui il 10% sono difettose. Dato un campione di 10 qual è la probabilità di avere (k) difettose

3

X = # lampadine difettose

$X \sim \text{Hyper}(100, 900, 10)$ $\qquad \mathbb{E}[X] = 1$

$Y \sim \text{Bin}(10, \frac{1}{10})$ $\qquad \mathbb{E}[Y] = 1$

k	$\mathbb{P}[X=k]$	$\mathbb{P}[Y=k]$
0	34.7 %	34.9 %
1	38.9 %	36.7 %
2	19.4 %	19.3 %
3	5.7 %	5.7 %
>3	1.2 %	1.3 %

ipergeometrica — binomiale

Esercizio

In un'urna ci sono R palline rosse e B blu

ne vengono estratte due senza rimpiazzo.

S è il numero di palline rosse estratte calcolare

$\mathbb{E}[S]$ e $\mathrm{VAR}[S]$

$$X = \begin{cases} 1 & \text{se prima pallina rossa} \\ 0 & \text{se prima pallina blu} \end{cases} \qquad Y = \begin{cases} 1 & \text{se prima pallina rossa} \\ 0 & \text{se prima pallina blu} \end{cases}$$

$X \sim B(p)$ $\quad p = \mathbb{P}[R_1] = \frac{R}{R+B}$ $\quad Y \sim B(q)$ $\quad q = \mathbb{P}[R_2] = \frac{(R+B-1)R}{(R+B)(R+B-1)} = \frac{R}{R+B}$

$X \sim Y$ $\quad S = X + Y$

$$\mathbb{E}[S] = \mathbb{E}[X+Y] = \mathbb{E}[X] + \mathbb{E}[Y] = 2\,\mathbb{E}[X] = 2p = 2\frac{R}{R+B}$$

($X \sim Y$, $X \sim B(p)$)

$$\mathrm{VAR}[S] = \mathrm{VAR}[X+Y] = \mathrm{VAR}[X] + \mathrm{VAR}[Y] + 2\,\mathrm{cov}[X,Y] =$$

$$= 2\frac{R}{R+B}\frac{B}{R+B} - 2\frac{R}{R+B}\frac{B}{R+B}\frac{1}{R+B-1} = 2\frac{R}{R+B}\frac{B}{R+B}\left(1 - \frac{1}{R+B-1}\right) = 2\frac{R}{R+B}\frac{B}{R+B}\frac{R+B-2}{R+B-1}$$

$$\mathrm{VAR}[X] = \mathrm{VAR}[Y] = p(1-p) = \frac{R}{R+B}\frac{B}{R+B} \qquad 1-p = 1-\frac{R}{R+B} = \frac{B}{R+B}$$

($X \sim Y$, $X \sim B(p)$)

$$\mathrm{cov}[XY] = \mathbb{E}[XY] - \mathbb{E}[X]\,\mathbb{E}[Y] = \frac{R(R-1)}{(R+B)(R+B-1)} - \left(\frac{R}{R+B}\right)^2 = \frac{R}{(R+B)}\,\frac{(R-1)(R+B) - R(R+B-1)}{(R+B-1)(R+B)}$$

$$= \frac{R}{R+B}\,\frac{R(R+B) - (R+B) - R(R+B) + R}{(R+B-1)(R+B)} = -\frac{R}{R+B}\frac{B}{(R+B)}\frac{1}{(R+B-1)}$$

$$\mathbb{E}[XY] = \mathbb{P}[X=1, Y=1] = \mathbb{P}[R_1 R_2] = \frac{R(R-1)}{(R+B)(R+B-1)} \qquad XY = \begin{cases} 1 & X=Y=1 \\ 0 & \text{altrimenti} \end{cases}$$

Esercizio (non fatto a lezione)

Due giocatori lanciano una moneta $\mathbb{P}[T] = p$

N e M volte rispettivamente.

Sapendo che sono uscite n teste

qual è la probabilità che il primo giocatore

abbia ottenuto k teste

$$\mathbb{P}[S_1 = k \mid S_1 + S_2 = n] = \frac{\mathbb{P}[S_1 = k]\,\mathbb{P}[S_2 = n-k]}{\mathbb{P}[S_1 + S_2 = n]}$$

$$= \frac{\binom{N}{k} p^k (1-p)^{N-k} \binom{M}{n-k} p^{n-k} (1-p)^{M-(n-k)}}{\binom{N+M}{n} p^n (1-p)^{M+N-n}}$$

$$= \frac{\binom{N}{k}\binom{M}{n-k}}{\binom{N+M}{n}}$$

$S_1 \sim \text{Bin}(N, p)$ $\quad S_2 \sim \text{Bin}(M, p)$ $\quad S_1 \perp\!\!\!\perp S_2$

$S_1 + S_2 \sim \text{Bin}(N+M, p)$

• UNIFORME (DISCRETA) •

$$X \sim \mathcal{U}(\{1,\dots,N\})$$

MODELLO: LANCIO UN DADO EQUILIBRATO CON N FACCE

PROPRIETÀ

a) $X \in \{1,\dots,N\}$

b) $\mathbb{P}[X = k] = \frac{1}{N}$

c) $\mathbb{E}[X] = \frac{N+1}{2}$ $\quad$ $\mathrm{VAR}[X] = \frac{N^2-1}{12}$

$$\mathbb{E}[X] = 1\,\frac{1}{N} + \dots + N\,\frac{1}{N} = \frac{1}{N}(1+\dots+N) = \frac{1}{N}\,\frac{N(N+1)}{2} = \frac{N+1}{2}$$

$$\mathrm{VAR}[X] = \mathbb{E}[X^2] - \mathbb{E}[X]^2 = \frac{(2N+1)(N+1)}{6} - \left(\frac{N+1}{2}\right)^2 = (N+1)\,\frac{2(2N+2) - 3(N+1)}{12} = \frac{N^2-1}{12}$$

$$\mathbb{E}[X^2] = 1\,\frac{1}{N} + \dots + N^2\,\frac{1}{N} = \frac{1}{N}(1+\dots+N^2) = \frac{1}{N}\,\frac{(2N+1)(N+1)N}{6} = \frac{(2N+1)(N+1)}{6}$$

Densità uniforme (continua)

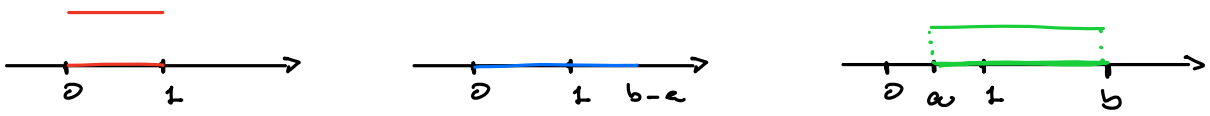
$$X \sim \mathcal{U}([a,b]) \qquad a < b$$

c) $$f(x) = \begin{cases} \frac{1}{b-a} & a \le x \le b \\ 0 & \text{altrove} \end{cases}$$

b) $\mathbb{E}[X] = \frac{a+b}{2}$ $\quad$ $\mathrm{VAR}[X] = \frac{(b-a)^2}{12}$

c) $$F(x) = \begin{cases} \frac{1}{b-a}(x-a) & a \le x \le b \\ 0 & \text{altrove} \end{cases}$$

d) $X = (b-a)\,X_0 + a$ dove $X_0 \sim \mathcal{U}([0,1])$



Densità esponenziale

$T \sim \mathcal{E}(\lambda)$ $\quad \lambda > 0$ $\quad \tau = \frac{1}{\lambda}$

- $f(t) = \begin{cases} \lambda e^{-\lambda t} & x \geq 0 \\ 0 & \text{altrove} \end{cases}$ $\quad = f(x) = \frac{1}{\tau} e^{-\frac{t}{\tau}}$

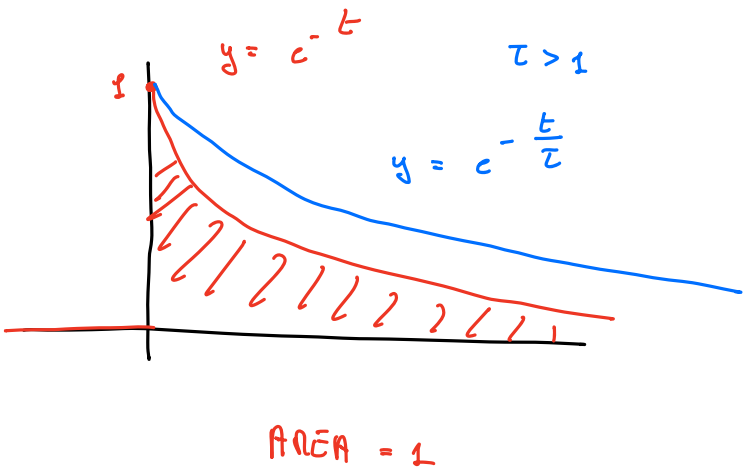
- $\mathbb{E}[T] = \frac{1}{\lambda} =$ $\quad VAR = [\ ] = \frac{1}{\lambda^2} = \tau^2$

- $F(t) = \begin{cases} 1 - e^{-\lambda t} & x \geq 0 \\ 0 & x < 0 \end{cases}$

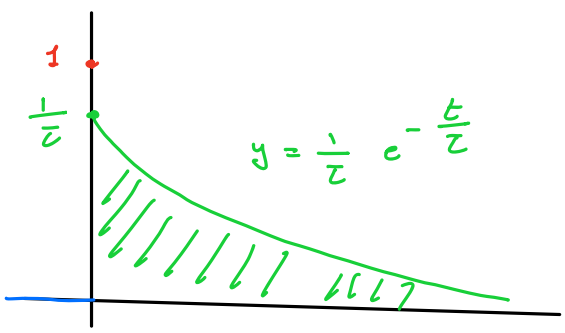
- $\mathbb{P}[T > t] = e^{-\lambda t} = e^{-\frac{t}{\tau}} \qquad t \geq 0$

- $\mathbb{P}[T > t+h \mid T > h] = \mathbb{P}[T > h]$ ASSENZA DI MEMORIA

$\forall\, t, h \geq 0$



AREA = 1



AREA = 1

Esercizio

Un call center risponde in media a 12 telefonate all'ora.

c) Qual è la probabilità di aspettare meno di 4 minuti?

b) Avendo aspettato 2 minuti, qual'è la probabilità di aspettare meno di 5 minuti?

Ipotesi T = tempo di attesa $T \sim \mathcal{E}(\lambda)$

$$\mathbb{E}[T] = \frac{1}{\lambda} = \frac{60}{12} = 5 \qquad \lambda = \frac{1}{5} = 0.2$$

$$\mathbb{P}[T \le 4] = 1 - \mathbb{P}[T > 4] = 1 - e^{-\lambda t} = 1 - e^{-\frac{4}{T_0}}$$

$$= 1 - e^{-\frac{4}{5}} \simeq 55\%$$

$$\mathbb{P}[T \le 5 \mid T > 2] = 1 - \mathbb{P}[T > 5 \mid T > 2]$$

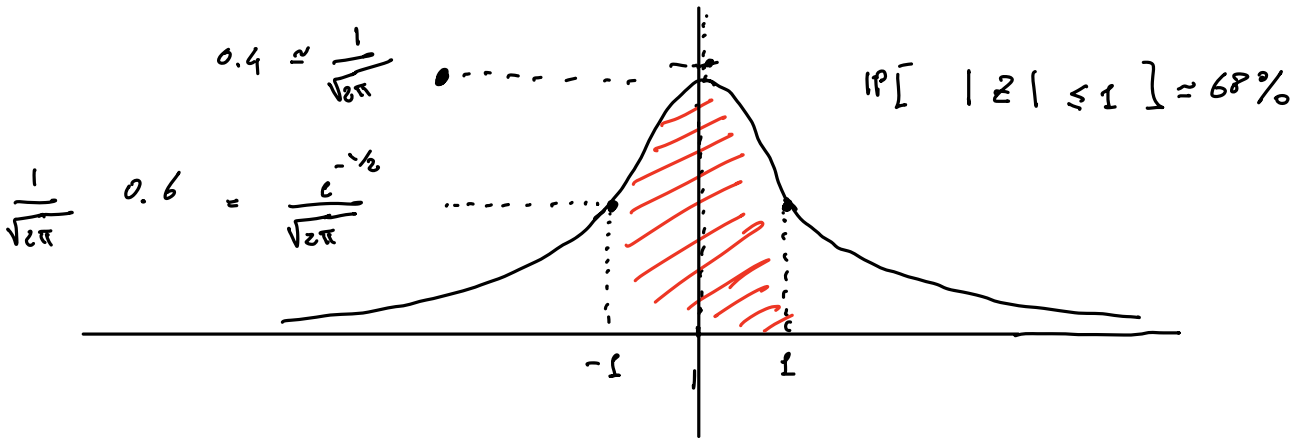
$$= 1 - \mathbb{P}[T > 3] = 1 - e^{-3/5} = 45\%$$

Lezione 30/04/26 2 ore

- Densità normale o gaussiana

a) normale standard $Z \sim N(0;1)$

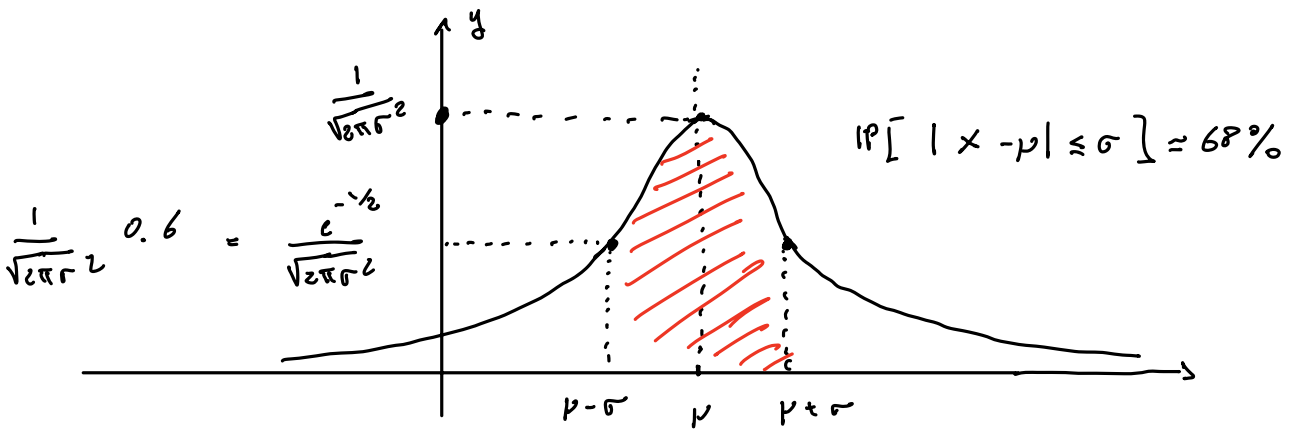
$$f_Z(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}}$$



b) normale di parametri $\mu \in \mathbb{R}$ e $\sigma > 0$

$$X \sim N(\mu, \sigma^2)$$

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \qquad x \in \mathbb{R}$$



Proprietà $\quad X \sim N(\mu, \sigma^2) \qquad Z \sim N(0, 1)$

a) $\quad X = \sigma Z + \mu$

b) $\quad E[Z] = 0 \qquad VAR[Z] = 1$

$\quad E[X] = \mu \qquad VAR[X] = \sigma^2$

c) $\quad$ dati $\quad a, b \in \mathbb{R} \quad a \neq 0 \quad$ allora

$\quad aX + b \sim N(a\mu + b, \; a^2\sigma^2)$

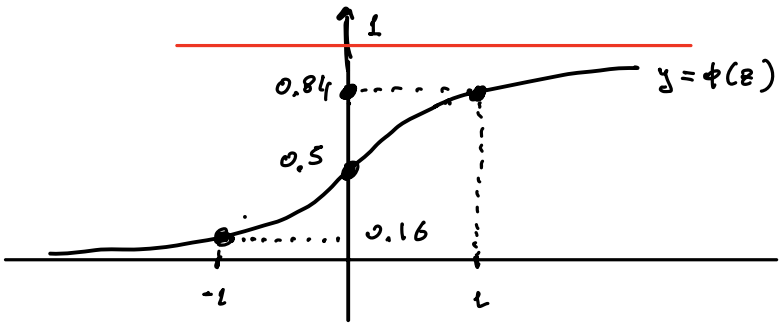
d) $\quad$ se $\quad X \sim N(\mu_1, \sigma_1^2)$ e $Y \sim N(\mu_2, \sigma_2^2) \quad X \perp\!\!\!\perp Y$

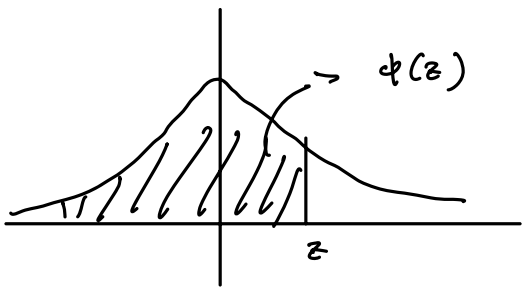
$\quad$ allora $\quad$ dati $\quad a, b \in \mathbb{R} \quad a^2 + b^2 \neq 0$

$$aX + bY \sim N(a\mu_1 + b\mu_2, \; a^2\sigma_1^2 + b^2\sigma_2^2)$$

- Funzione di distribuzione cumulata normale

$$\Phi(z) = P[Z \le z] = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{z} e^{-\frac{t^2}{2}} dt = \frac{1}{2}\left[1 + \operatorname{erf}\left(\frac{z}{\sqrt{2}}\right)\right]$$

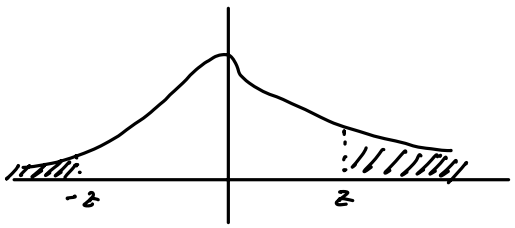




- Proprietà

a) $\Phi(0) = 1/2$

b) $\Phi(z) + \Phi(-z) = 1$



c) $P[|z| \le \varepsilon] = [\varepsilon < |z|] = 2(\Phi(\varepsilon) - 1) = 2(\Phi(\varepsilon) - \Phi(-\varepsilon))$ $\quad \varepsilon > 0$

d) $P[|z| > \varepsilon] = 1 - P[|z| \le \varepsilon] = 1 - 2\Phi(-\varepsilon) = 1 - 2(1 - \Phi(\varepsilon))$

e) $\Phi(-z) = 1 - \Phi(z) \le \frac{1}{\sqrt{2\pi}} \frac{1}{z} e^{-\frac{z^2}{2}}$ $\quad z > 0$

z	$\Phi(-z)$	approx.
1	16 %	30 %
2	2.3 %	2.7 %
3	0.13 %	0.15 %
4	0.0032 %	0.0034 %

Dim: $\Phi(-z) = P[Z \ge z] = \frac{1}{\sqrt{2\pi}} \int_{z}^{+\infty} e^{-\frac{t^2}{2}} dt \le \frac{1}{\sqrt{2\pi}} \int_{z}^{+\infty} \frac{t}{z} e^{-\frac{t^2}{2}} dt$

f) $X \sim N(\mu, \sigma^2)$ $\quad X = \sigma Z + \mu$ $\quad Z \sim N(0,1)$

Dati $a, b \in \mathbb{R}$ $\quad a < b$

$$\mathbb{P}[a \le X \le b] = \mathbb{P}\left[\frac{a-\mu}{\sigma} \le \underbrace{\frac{X-\mu}{\sigma}}_{Z} \le \frac{b-\mu}{\sigma}\right] = \Phi\left(\frac{b-\mu}{\sigma}\right) - \Phi\left(\frac{a-\mu}{\sigma}\right)$$

- Uso delle tabelle

	0.00	....	0.04
0,0	0.5		0.5160
....			
1.2	0.8849		0.8925
⋮			
-1.2	0.1151		0.1075

Es.:

$X \sim N(1, 0.16)$ $\quad \mu = 1$ $\quad \sigma^2 = 0.16$ $\quad \sigma = 0.4$

$$\mathbb{P}[0.6 \le X \le 1.3] = \mathbb{P}\left[\frac{0.6-1}{0.4} \le \underbrace{\frac{X-1}{0.4}}_{Z} \le \frac{1.3-1}{0.4}\right]$$

$$= \mathbb{P}\left[-1 \le Z \le \frac{3}{4}\right] = \Phi(0.75) - \Phi(-1)$$

$$= 0.7734 - 0.1587 = 0.6147 \simeq 61.47 = 61\%$$

$$\mathbb{P}[X > 2] = \mathbb{P}\left[\frac{X-1}{0.4} > \frac{2-1}{0.4}\right] = \mathbb{P}[Z > 2.5] =$$

$$= 1 - \Phi(2.5) = \Phi(-2.5) = 0.0062 \simeq 0.6\%$$

$$1 - 0.9938$$